Week 3 Lab Meeting

Summer 2026

Deviations

- What happened this week?
- What did you do?
- Any suggestions to improve the system?

Other logistics

What did we talk about last week?

- Article structure
- Issues with previous APWs studies (e.g., small samples, big claims)
- Heterogeniety
- Different operational definition for work performance

Why do we choose TSST?
How does TSST create stress?
What do we measure?
Why might people respond differently?

Walk and Talk! Come back after 5 minutes

Why do we choose TSST?

- Controlled and standardized protocol
- Strong enough
- Similar to work stressors

How does TSST create stress?

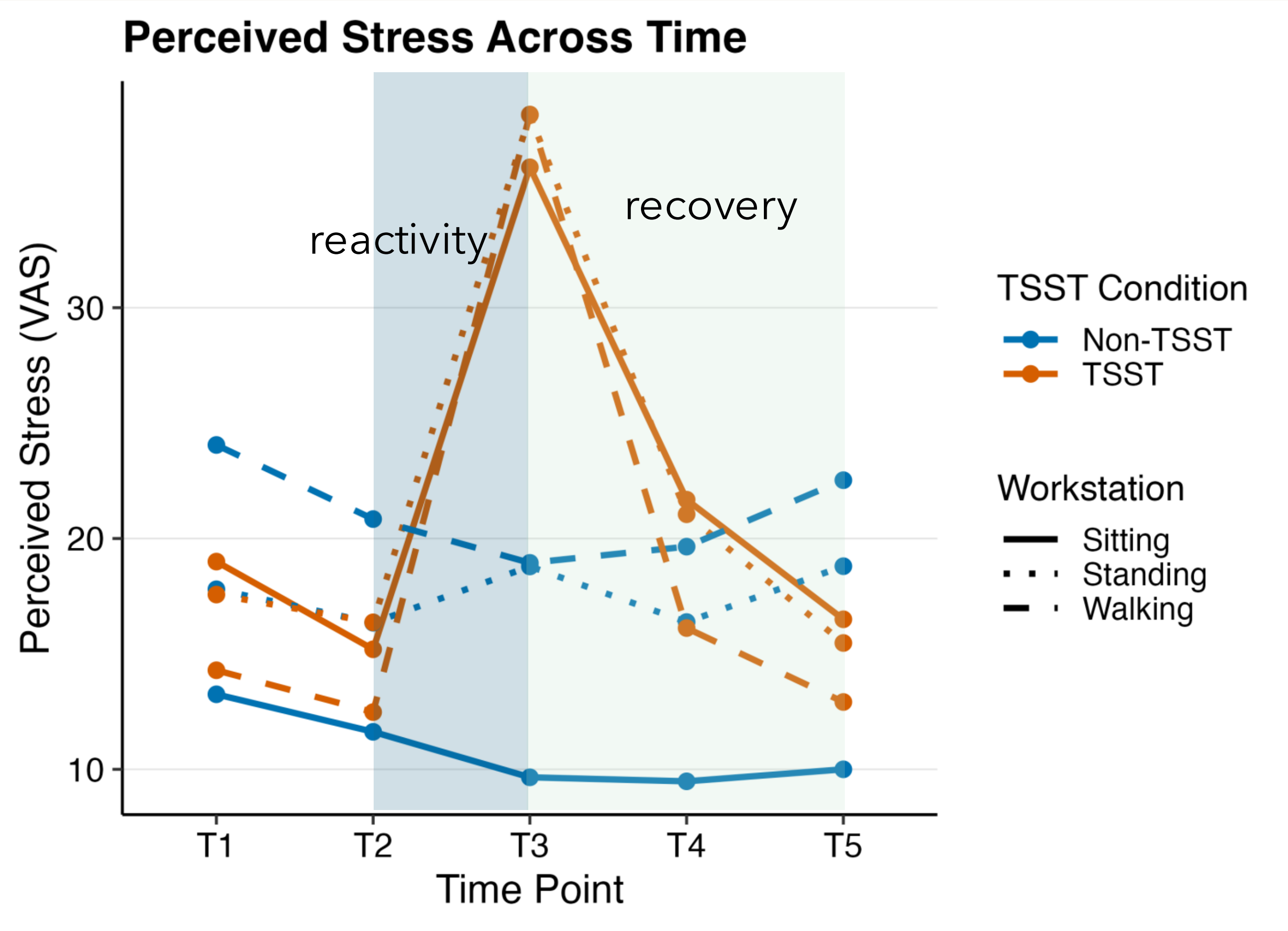
- Social evaluative threat
- Uncontrollability

What do we measure?

- Subjective experience
- Mood/affect
- Cardiovascular response
- HPA Axis

Why might people respond differently?

- **Biological factors**
  - *age*, sex/gender, hormones, menstrual cycle, oral contraceptives
- **Psychological factors**
  - *personality, appraisal, perceived control, social status*
- **Health/history factors**
  - *chronic stress, early life stress, mental health conditions*
- **Genetic/epigenetic factors**
  - *inherited differences + experience-shaped biology*
- **Contextual factors**
  - *stigma, identity-related threat, prior TSST exposure*



Perceived Stress Across Time
reactivity
recovery
Perceived Stress (VAS)
30
20
10
T1
T2
T3
T4
T5
Time Point
TSST Condition
Non-TSST
TSST
Workstation
Sitting
Standing
Walking

Psychological stress measures

- VAS stress: “How stressed do you feel?”
- PANAS: positive and negative affect
- Fast, repeatable, participant-centered

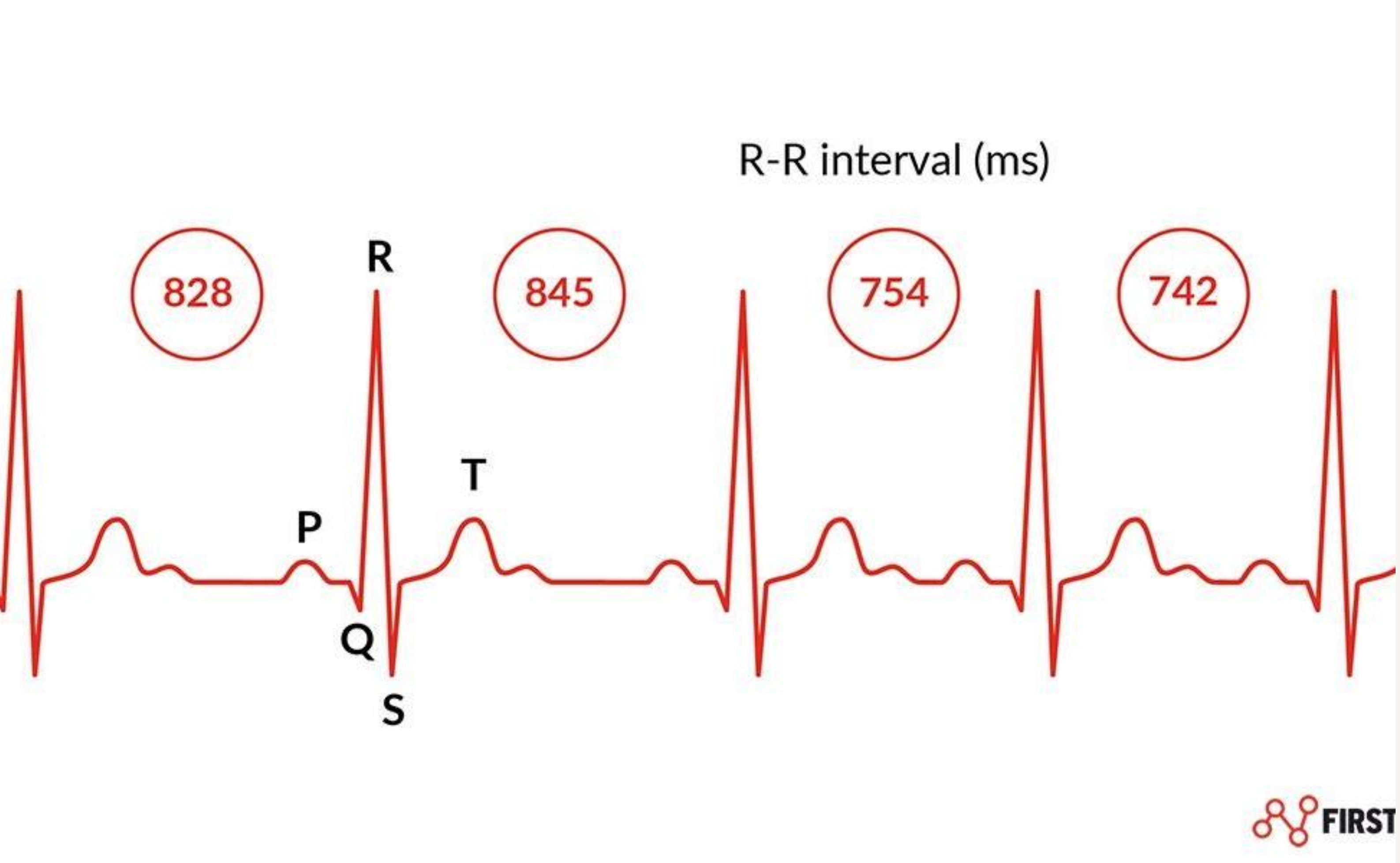
Salivary cortisol

- Marker of HPA-axis activation
- Slower than HR
- Often peaks after stressor
- Sensitive to outside influences

Heart rate: the fast responder

- Beats per minute
- Responds quickly
- Captures arousal/activation
- Easy to interpret, but not stress-specific (walking raises HR as well)

Heart rate variability



- HRV = variation between heartbeats
- Higher HRV often reflects better regulation
- Stress often lowers HRV
- Movement/artifact can affect HRV

HRV time-domain measures

- Actual time gaps between heartbeats
- **RMSSD** = **Root Mean Square of Successive Differences**
  - *Captures beat-to-beat variability between normal heartbeats*
  - *Often interpreted as parasympathetic/vagal regulation*
- **SDNN** = **Standard Deviation of Normal-to-Normal Intervals**
  - *Captures overall variability in heartbeat intervals*
  - *Interpretation depends on recording length*

HRV frequency-domain measures

- breaks HRV into rhythm bands
- **HF Power = High frequency power**
  - *Closely tied to breathing and parasympathetic/vagal activity*
- **LF Power = Low prequency power**
  - *A mix of sympathetic activity, parasympathetic activity, and baroreflex regulation, which is the body's blood-pressure control system*

How can we reduce the "noise"

- **If people differ so much, how can we still test whether treadmill walking helps stress?**
- Random assignment
- Repeated measures
- Multiple outcomes
- Health and demographic questions

Share your evaluator experience!